

Michael Marchev

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EDUCATION

Northeastern University, College of Engineering

Boston, MA

Candidate for BS in Mechanical Engineering and Design | GPA: 3.35

Dec 2028

Coursework: FEA & Design, Dynamics, Mechanics of Materials, Thermodynamics, Statics, Materials Science, Physics 2, Calculus 3, Differential Equations and Linear Algebra, Design Perspectives + Process

PROFESSIONAL EXPERIENCE

Stryker (Endoscopy)

San Jose, CA

R&D Mechanical Engineering Co-op

Jan 2026 - Jun 2026

- Designed **10 μ m resolution** optics bench with **4 motorized + 4 manual axes** and full blackout cage, including supplier management, machining + anodization, build, testing, and control documentation
- Created custom FDM/milled fixturing to integrate prototypes with HALT + linear/torsional Instrons
- Led Verification & Validation on next-gen endoscopy camera heads to establish patient/user safety, lifecycle FOS, and support FDA compliance: vibration, thermal, thermopile, mech. reliability testing
- Optimized prototype manufacturing methods (MJP to FDM) to reduce materials **spending by 57%**

Richey & Clapper Inc.

Sudbury, MA

Power Equipment Mechanic (STIHL Certified Silver Technician)

May 2025 - Aug 2025

- Fully rebuilt 2 & 4 stroke petrol/diesel/electric motors, transmissions, gearboxes, carburetors, and serviced belts, hydraulics, and tires to support full range of contractor-grade lawn machinery
- Performed cost & failure mode analysis to determine repair and parts modification procedures

PROJECTS & LEADERSHIP

Generate Product Development Studio

Boston, MA

Mechanical Technical Lead - Automated X-Ray Survey Machine

Jul 2026 - Present

- Design top-level architecture of **~200 ft³ 5-axis** motorized gantry for detecting and logging errant X-ray emissions of industrial CT scanners, projected to **save ~1 hr** operator time per unit
- Manage **4 mechanical engineers**, direct subsystem design, assign action items, and run meetings
- Control frame/axis squareness and eliminate play in mounting for **projected 2-5mm repeatability**

Mechanical Engineer - Sageware

Sep 2025 - Dec 2025

- Automated reliable fabric upcycling through compact, process-based bead production solution
- Formulated robust cutting process with machined steel stamp and SLA resin fixture, linearly actuated by stepper motor, ball screw, and linear rail integrated with molding + dispensing systems

Mechanical Engineer - Uplift Solutions

Dec 2024 - Apr 2025

- Iteratively developed static & dynamic stiff conduit linkages to withstand axial load and fully constrain relative motion on powered walker with raising/tilting seat and legs

Northeastern University Center for STEM Education - Young Scholars' Program

Boston, MA

Program Co-Coordinator

Jun 2024 - Aug 2024

- Managed 6-week NSF-funded university research program for 27 high school seniors
- Coordinated experiential education, tutored on professional development/communication skills

Research Assistant & **Co-author - *Annals of Biomedical Engineering* (Vol. 53)**

Jun 2023 - Aug 2023

- Researched impact mechanics and force propagation in advanced combat helmet systems
- Quantified precompression of helmet padding via experimental testing (**89 subjects**) for FEA

VolunTeachable.com

Boston, MA

Co-Founder

Sep 2020 - Aug 2024

- Founded global linguistic exchange program for virtual mentorship of underprivileged children
- Organized relationships with **500+ students** and schools across South America, Asia, and Europe

CERTIFICATIONS & SKILLS

CAD/Manufacturing: Creo, SOLIDWORKS(CSWA, AM, Sustainability), Onshape, AutoCAD, FDM + SLA + MJP 3D Printing, Manual Mill, Soldering, Advanced Hand Tools, **Lean Six Sigma Green Belt**

Skills: Part Design, Product Development, DFM + DFA, C++, Python, MATLAB, InDesign, Photoshop

Languages: English: *Native*, Bulgarian: *Fluent*, Mandarin: *Conversational*

Interests: Espresso, Judo, Audio, Music, Hiking, Skiing, Running, Violin, Piano, Soccer